CYBERSECURITY
PROJECT REPORT

OWASP Juice Shop Web Application Security Assessment

A practical learning project focused on application exploration, HTTP traffic review, and selected security challenge scenarios.

Environment
Kali Linux virtual machine · OWASP Juice Shop running locally · Burp Suite Community Edition

Evidence captured
October 2026

Repository
github.com/gkmohomed/owasp-juice-shop-security-assessment

PROJECT SUMMARY

This report records the work evidenced in the supplied screenshots. It is intended as a portfolio learning record, not as a formal penetration-test certification or an assessment of a production system.

1. Project Overview

This project used OWASP Juice Shop, an intentionally vulnerable web application designed for security education. The work focused on exploring the local application, reviewing browser and proxy traffic, inspecting the application site map, and completing selected practice challenges.

The supplied evidence contains six successful challenge notices across the captured sessions: the Score Board discovery challenge and five additional scenarios - DOM XSS, Login Admin, Admin Section, Zero Stars, and View Basket. The Score Board screenshot itself records 1 of 186 challenges solved at that point in time; the five later success banners provide evidence of additional completions.

Objectives

- Become familiar with the Juice Shop interface and its local application routes.
- Observe baseline browser network activity generated by the application.
- Use Burp Suite to review HTTP history, selected requests, responses, and the target site map.
- Complete selected built-in practice challenges and retain screenshot evidence.

Scope and environment

Item	Evidence-based description
Application	OWASP Juice Shop training application.
Target	Local loopback address shown in the screenshots: 127.0.0.1:3000.
Operating environment	Kali Linux virtual machine shown running in VMware Workstation.
Traffic inspection	Burp Suite Community Edition; browser developer tools/network panel.
Evidence	11 screenshots supplied by the project author.

Tools observed in the evidence

OWASP Juice Shop, Burp Suite Community Edition, a web browser, Kali Linux, and VMware Workstation. No external production host is within the documented scope.

Approach

The workflow represented by the evidence was: (1) open the local training application; (2) observe baseline browser network activity; (3) inspect HTTP traffic and the target site map in Burp Suite; (4) work through selected Juice Shop challenges; and (5) capture application success notices and supporting screenshots.

2. Application Exploration and Baseline Review

The initial screenshots establish that Juice Shop was accessed locally and that browser network activity was observed while the application loaded.

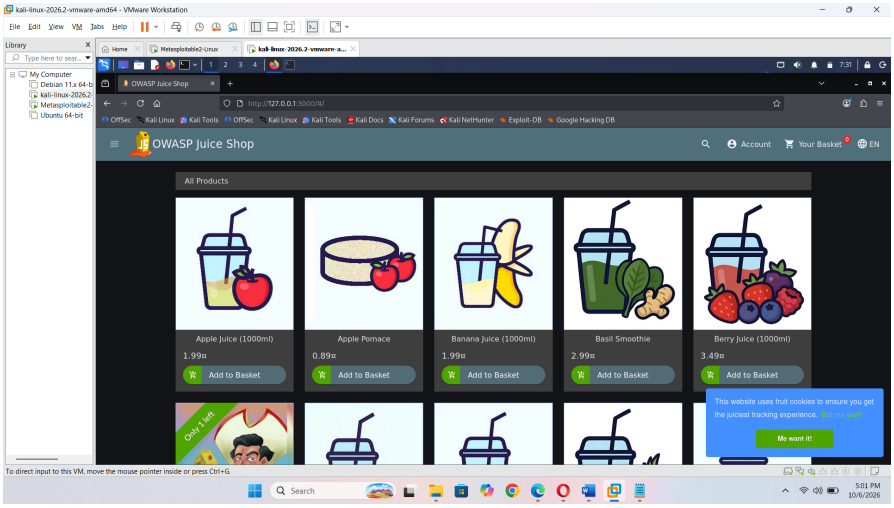


Figure 1. Juice Shop home page showing the product catalogue in the local training environment.

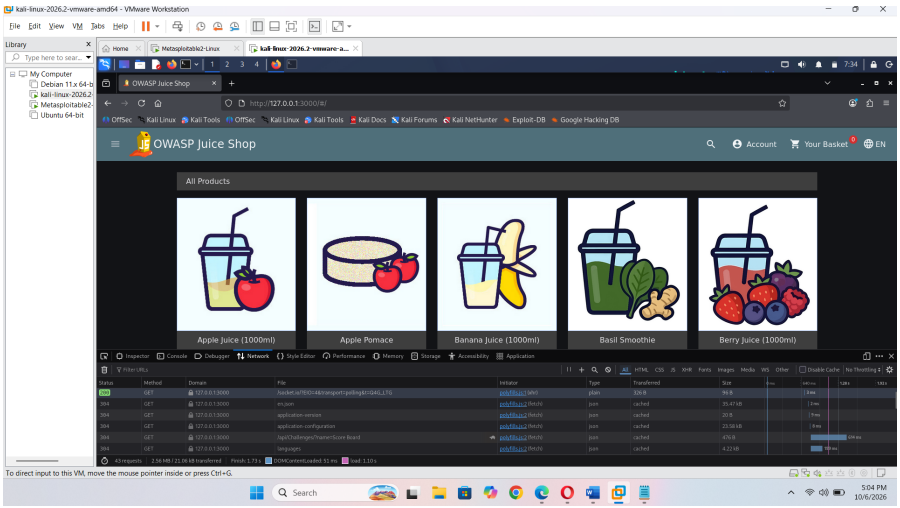


Figure 2. Browser developer tools Network panel showing requests generated during application loading. The capture provides a baseline view of local client-server activity.

Observation

The application was reachable through the local loopback interface. The network panel showed requests to the local Juice Shop host, giving a starting point for subsequent traffic review.

3. HTTP Traffic and Application Mapping

Burp Suite was used as an intercepting proxy to review application traffic. The submitted evidence includes an HTTP history view, a target site map, and an individual request/response inspection.

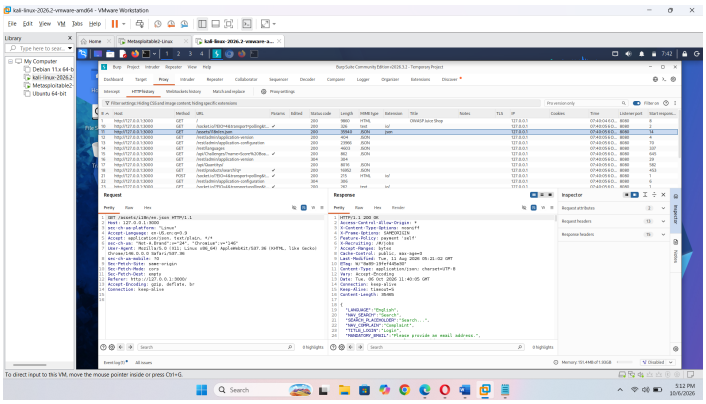


Figure 3. Burp Suite Proxy HTTP history showing requests to the local Juice Shop application and the corresponding response metadata.

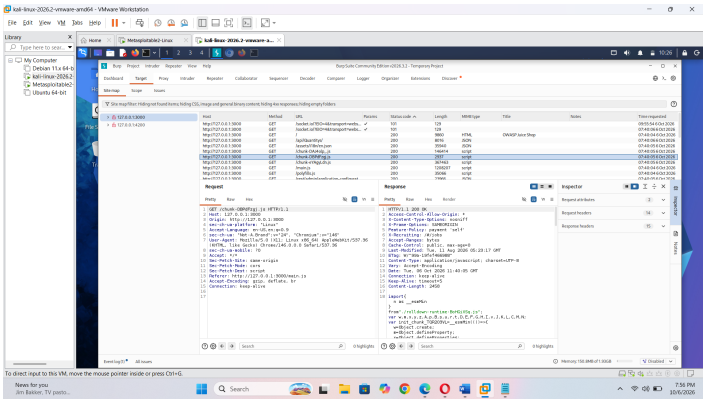


Figure 4. Burp Suite Target site map displaying discovered local hosts and application resources, with a selected request and response.

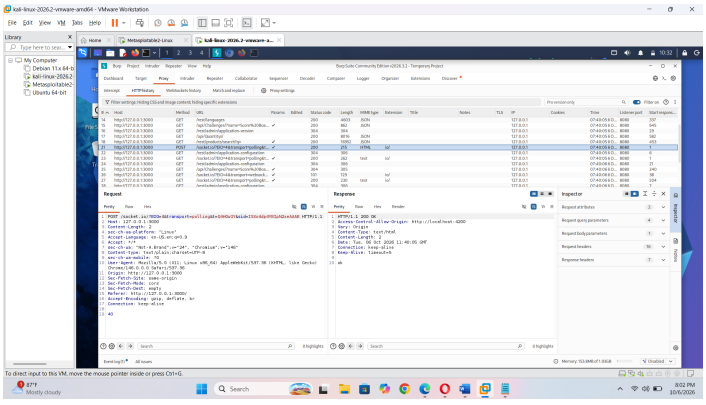


Figure 5. Selected local Socket.IO polling request and response in Burp Suite. Although the screenshot filename refers to a login request, the visible entry is a Socket.IO polling exchange rather than a credential submission.

4. Challenge Evidence - Discovery and DOM XSS

The next screenshots contain green in-application success notices. These notices are the evidence used to identify completed challenges; challenge names are retained as shown in the Juice Shop interface.

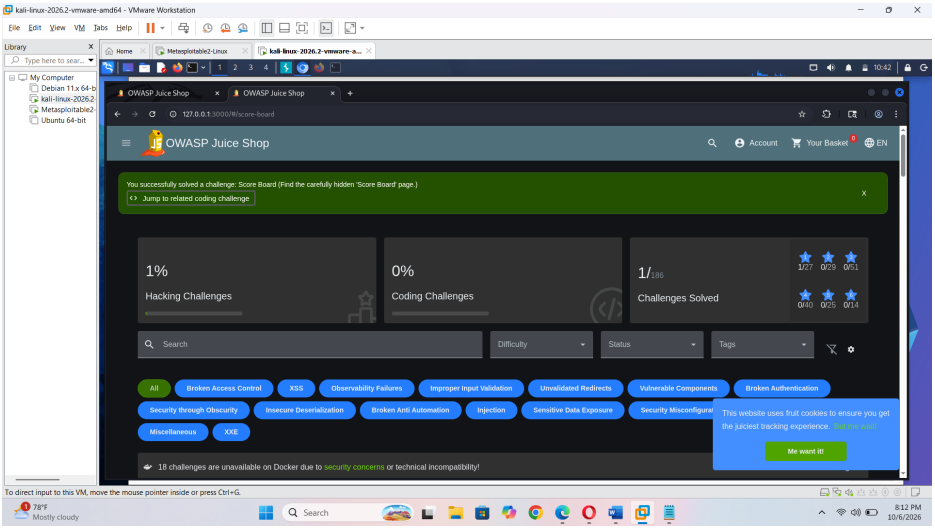


Figure 6. Score Board discovery challenge success notice. The same capture shows a scoreboard progress value of 1/186 at the time this screenshot was taken.

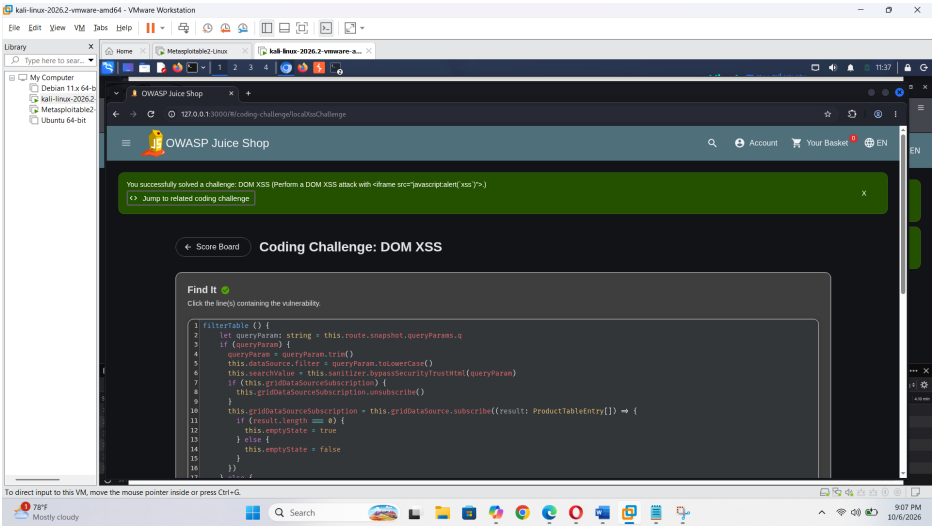


Figure 7. DOM XSS challenge success notice displayed by Juice Shop. The screenshot also shows the associated coding-challenge view.

5. Challenge Evidence - Access Scenarios

The two screenshots below show successful challenge notices for the Login Admin and Admin Section scenarios. They document outcomes in the intentionally vulnerable training application only.

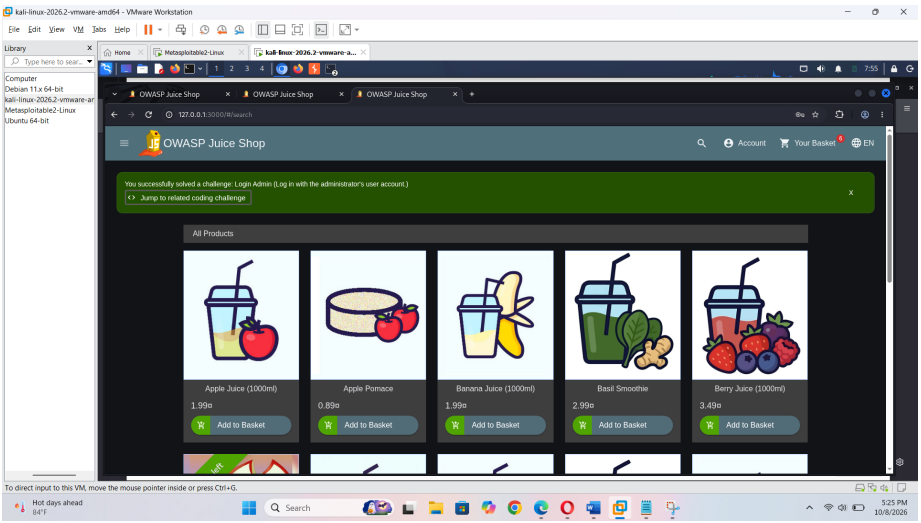


Figure 8. Login Admin challenge success notice displayed after the practice scenario was completed.

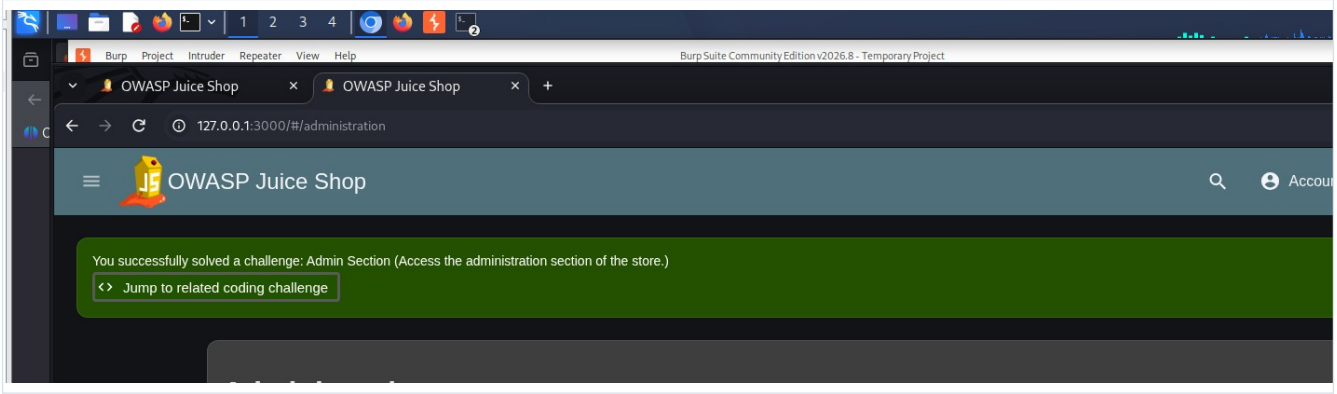


Figure 9. Admin Section challenge success notice and administration page heading. The evidence image is cropped to avoid reproducing the user and feedback lists shown further down the original page.

6. Challenge Evidence - Application Functionality

These captures show the in-application success notices for the Zero Stars and View Basket scenarios.

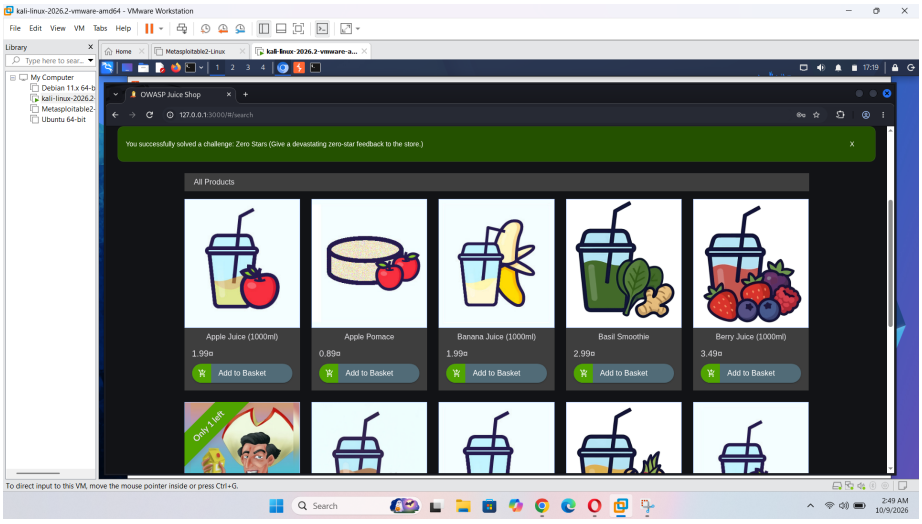


Figure 10. Zero Stars challenge success notice displayed in the Juice Shop product catalogue.

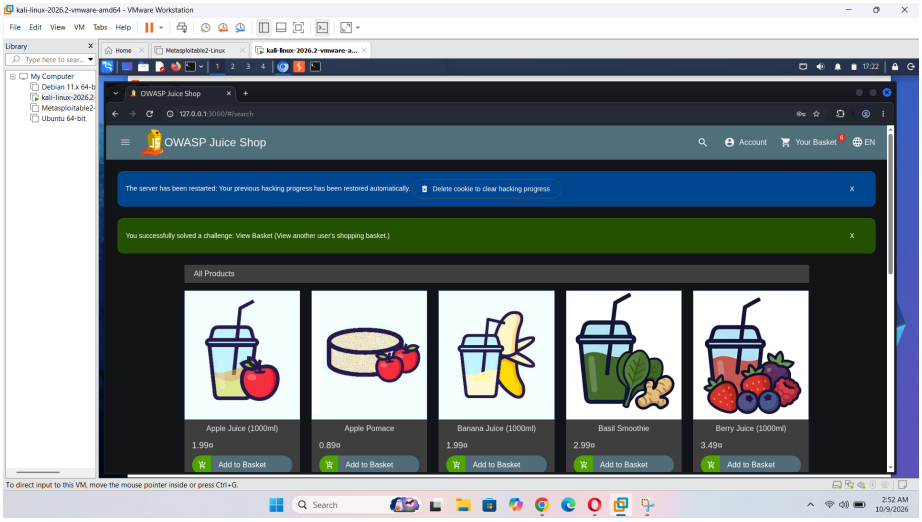


Figure 11. View Basket challenge success notice. A separate informational banner indicates that the training application restored prior challenge progress after a server restart.

7. Results, Learning Outcomes, and Conclusion

Challenge evidence summary

Challenge	What the supplied screenshot evidences
Score Board	Success notice for finding the hidden Score Board page.
DOM XSS	Success notice for the DOM XSS practice challenge.
Login Admin	Success notice for the Login Admin practice challenge.
Admin Section	Success notice for accessing the administration section in the lab.
Zero Stars	Success notice for the zero-star feedback challenge.
View Basket	Success notice for the View Basket challenge.

Key learning outcomes

- Observed local web application traffic using browser developer tools and Burp Suite.
- Reviewed HTTP history and application resources through the Burp Suite site map.
- Practised selected security and application-functionality challenges in a purpose-built training application.
- Collected and organized screenshot evidence to support technical documentation.

Limitations

The screenshots provide evidence of selected activities and challenge completion notices. They do not constitute a comprehensive security assessment, an independent validation of every application component, or a formal risk rating. Detailed exploit payloads, credential material, and instructions for reproducing access-control bypasses are intentionally omitted from this public portfolio report.

Ethical scope

All documented activity was performed against the local OWASP Juice Shop training instance shown at 127.0.0.1:3000. These techniques must only be practised in environments owned by the learner or where explicit authorization has been provided.

Conclusion

This project provided practical exposure to web application exploration, HTTP traffic inspection, application mapping, and selected security challenge scenarios. The evidence has been organized as a learning record and portfolio item to support continued study of web application security.

Project repository
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